KOLITHA WARNAKULASOORIYA

AI Engineer | Machine Learning Engineer | Applied AI Engineer

Mobile, AL
251.373.5542 | kolitha.warnakulasooriya@outlook.com | linkedin.com/in/kolitha-warnakulasooriya | github.com/kolithawarnakulasooriya

Professional Summary

AI/ML Engineer and Software Engineer with 7+ years of experience spanning artificial intelligence, machine learning research, production software engineering, distributed systems, and autonomous robotics. Ph.D. in Computing with hands-on experience developing LLM-enabled applications, ML experimentation pipelines, cloud-native microservices, and multi-agent autonomous systems using Python, Java, TypeScript, PyTorch, FastAPI, Docker, Kubernetes, AWS, MLflow, and ZenML. Delivered measurable improvements including 67% better optimization performance, 25% higher autonomous-system completion rates, 15–20% improved prediction performance, and 90%+ production test coverage. Experienced in translating research prototypes into tested, scalable engineering systems through API development, experimentation, deployment, monitoring, system design, and cross-functional collaboration.

Technical Skills

Programming: Python, Java, C++, TypeScript, JavaScript, SQL
AI / Machine Learning: PyTorch, Hugging Face Transformers, Machine Learning, Deep Learning, Large Language Models (LLMs), Generative AI, Reinforcement Learning, Computer Vision, Swarm Intelligence, Multi-Agent Systems, Model Evaluation, Optimization
LLM / AI Engineering: OpenAI-compatible APIs, Hugging Face Tokenizers, Accelerate, PEFT, LoRA, vLLM, pgvector, AWQ, Tool Calling, MCP Interfaces, State Machines
MLOps / Cloud / DevOps: MLflow, ZenML, FastAPI, Docker, Kubernetes, AWS, Google Cloud, NVIDIA Container Toolkit, Jenkins, Git, CI/CD, Linux, Datadog
Backend / Data: REST APIs, GraphQL, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, Redis, AWS Lambda, API Gateway
Robotics: ROS 2, rclpy, rclcpp, DDS, Gazebo, UAV Systems, Distributed Robotics, Robotic Perception

Professional Experience

University of South Alabama **Mobile, AL**
Visiting Assistant Professor *May 2025 – Present*

- Conduct applied research in AI/ML, autonomous systems, robotic perception, and multi-agent robotics while advising 2 senior design teams developing UAV navigation, path-planning, and coordinated autonomous-system solutions.
- Translate autonomous-system research problems into system architectures, experiments, simulation workflows, and physical validation strategies for robotics and edge-AI applications.
- Teach robotics, computer architecture, and digital logic systems across 200-, 400-, and 500-level engineering courses.
- Developed 4 research proposals in agentic and autonomous robotics by defining research objectives, technical architectures, experimental methods, and broader impacts for NSF-, Amazon-, and Google-related opportunities.

Research Assistant *Jan 2022 – May 2025*

- Engineered a distributed robotics framework coordinating **15 heterogeneous UAVs** using leader selection, formation control, real-time sensor streams, communications, and multi-agent decision-making, validating performance through **42+ physical experiments**.
- Increased autonomous mission completion rates by **25% across 3 UAV research initiatives** by leading simulation, algorithm implementation, experimental design, physical testing, field validation, and modular system integration using Docker/Kubernetes-based services and APIs.
- Improved convergence speed and resource utilization by **67% versus 10+ optimization algorithms across 62 benchmark tests and 14 experiments** by developing and evaluating a forest-inspired swarm-intelligence optimization algorithm.
- Built reproducible ML experimentation and analysis workflows using Python, MLflow, and ZenML to track experiments, process metrics, evaluate algorithm performance, and support repeatable research.
- Authored peer-reviewed research on autonomous systems, distributed decision-making, optimization, and robotic intelligence, contributing to a doctoral research program resulting in six first-author publications.

SyscoLabs Technologies **Colombo, Sri Lanka**
Lead Engineer (Associate) *Aug 2021 – Dec 2021*

- Led a cross-functional team of **10+ software, QA, product, and DevOps engineers** delivering 6 engineering initiatives while reducing administrative overhead by **50%** through improved technical planning and Agile execution.
- Led system design, sprint planning, code reviews, engineering-quality practices, and cross-team technical coordination for production software.

- Maintained **zero downtime for critical business services** during high-priority production incidents by coordinating diagnosis, troubleshooting, recovery, and cross-functional incident response.

Senior Software Engineer *Dec 2019 – Aug 2021*

- Improved arrival-time prediction performance by **15–20%** through statistical analysis and algorithmic refinements implemented in Python and Java.
- Achieved **90%+ test coverage** for a restructured production BFF microservice by implementing unit and integration testing and integrating Docker/Kubernetes deployment workflows.
- Designed and developed scalable cloud applications using AWS, Spring Boot, Node.js, React/Redux, microservices, REST APIs, automated testing, predictive analytics, and CI/CD pipelines.

Software Engineer *Jul 2018 – Nov 2019*

- Accelerated ML and backend experimental workflows by **40%** by developing an open-source in-memory mock API service for rapid testing and application development.
- Developed distributed web and backend services using Spring Boot, Express.js, REST, GraphQL, WebSockets, AWS Lambda, API Gateway, React, and React Native within AWS microservice environments.

Selected AI / ML Projects

CODEX-Autocomplete – LLM-Powered VS Code Extension **2026**

github.com/kolithawarnakulasooriya/CODEX-Autocomplete

TypeScript, OpenAI Responses API, OAuth PKCE, Streaming/SSE, VS Code API

- Built an LLM-powered VS Code extension that generates cursor-aware inline code completions using streaming OpenAI Responses API requests and incremental SSE processing.
- Engineered production-oriented request handling with cancellation, debounce, timeout, retry, local rate limiting, caching, bounded context generation, and completion post-processing.
- Implemented secure ChatGPT OAuth PKCE authentication, API-key fallback, encrypted VS Code SecretStorage, token refresh, unit tests, and integration testing.

E2E Machine Learning Framework **2025**

github.com/kolithawarnakulasooriya/E2E-ml-flow

Python, ZenML, MLflow, MLOps, Testing

- Developed an end-to-end ML framework supporting data ingestion, experimentation, model tracking, artifact management, and deployment workflows using ZenML and MLflow.
- Integrated experiment tracking, configurable ML stacks, model-deployment components, artifact stores, and automated unit-test coverage to improve experiment reproducibility.

AI-Driven Cyber-Physical System / Sensor Simulation Platform **2026**

github.com/kolithawarnakulasooriya/AI-driven-cyber-physical-system

Python, FastAPI, AsyncIO, WebSockets, MQTT, REST APIs

- Developed a real-time sensor simulation platform using FastAPI and AsyncIO with REST APIs, WebSocket telemetry, MQTT integration, configurable sensor models, and asynchronous data recording.
- Designed modular sensor-generation and integration components for AI-enabled cyber-physical-system experimentation and robotic perception workflows.

Vision Compass for Drones – Vision-Based UAV Localization **Sep 2025 – Feb 2026**

IEEE ICIPRoB 2026

Computer Vision, Vision Models, Visual Localization, Robotic Perception

- Developed and experimentally evaluated a vision-based localization methodology that uses characteristics of vision models to derive navigation information for autonomous UAV systems.
- Connected computer-vision model behavior with robotic localization and navigation, resulting in peer-reviewed publication at IEEE ICIPRoB 2026.

Education

University of South Alabama **Mobile, AL**

Ph.D. in Computing / Computer and Information Sciences *2025*

- **Dissertation:** Choosing Robust Leadership: Encompassing the Best-of-N Model and Swarm Intelligence Optimization for Heterogeneous Multiple Autonomous Unmanned Aerial Vehicle Systems.

University of Moratuwa **Colombo, Sri Lanka**

B.Sc. (Hons.) in Information Technology *2018*

Research

Six first-author peer-reviewed publications in autonomous systems, swarm intelligence, optimization, UAV systems, robotic perception, and multi-agent decision-making.